



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

M.TECH (POWER SYSTEM & AUTOMATION)

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

(With effect from 2019-2020 Admitted Batch onwards)

I-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Internal Marks	External Marks	Total Marks
M19 PS 1101	Advanced Power system operation and control	PC	3	3	0	0	25	75	100
M19 PS 1102	Power system dynamics and stability	PC	3	3	0	0	25	75	100
#PE-I	Program Elective-I	PE	3	3	0	0	25	75	100
#PE-II	Program Elective-II	PE	3	3	0	0	25	75	100
M19 RD 1101	Research Methodology and IPR	CC	2	2	0	0	25	75	100
M19 PS 1109	Power System Simulation Laboratory-I	PC	2	0	0	4	25	75	100
M19 PS 1110	Power system Laboratory	PC	2	0	0	4	25	75	100
#AC-1	Audit course -1	AC	0	2	0	0	0	0	0
Total			18	16	0	8	175	525	700

#PE-I	M19 PS1103	Electric Power Distribution System
	M19 PS1104	HVDC Transmission
	M19 PS1105	EHVAC Transmission
#PE-II	M19 PS1106	Artificial Intelligence Techniques
	M19 PS1107	Optimization Techniques
	M19 PS1108	Advanced Digital Signal Processing
#AC-1 & 2	M19AC0001	English for Research Paper Writing
	M19AC0002	Disaster Management
	M19AC0003	Sanskrit for Technical Knowledge
	M19AC0004	Value Education
	M19AC0005	Constitution of India
	M19AC0006	Pedagogy Studies
	M19AC0007	Stress Management by Yoga
	M19AC0008	Personality Development through Life Enlightenment Skills.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1101	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED POWER SYSTEM OPERATION & CONTROL								
Course Objectives: The main objectives of this subject are:								
1	To study the effect of generation with limited energy supply.							
2	To understand the economics of power system operation with thermal and hydro units.							
3	To study the unit commitment problem for economic load dispatch.							
4	To study the optimal power flow problem with solution.							
5	To study the load frequency control of two area systems with and without control.							
6	To study the effectiveness of interchange evaluation in interconnected power systems							
Course Outcomes								
S.No								Knowledge Level
1	Understand the effect of generation with limited energy supply.							K2
2	Solve generation dispatching scheme for thermal and hydro units.							K4
3	Solve the unit commitment problem for economic load dispatch.							K3
4	Get knowledge on load frequency control of single area and two area systems with and without control.							K2
5	Solve the interchange evaluation in interconnected power systems.							K3
6	Solve the optimal power flow problems with different solutions							K3
SYLLABUS								
UNIT-I (10 Hrs)	Generation with limited Energy supply: Take-or-pay fuel supply contract, composite generation production cost function. Solution by gradient search techniques, Base point and participation factor method, hard limits and slack variables, Fuel scheduling by linear programming. Hydroelectric plant models –short term hydrothermal scheduling problem – Gradient approach.							
UNIT-II (10 Hrs.)	Unit commitment problem: Constraints in UCP,UC solutions. UC Methods-priority list method, Forward Dynamic programming Approach and Lagrange Relaxation method.							
UNIT-III (10 Hrs.)	Optimal power flow: Solution of OPF, gradient method, Newton’s method, linear programming method with only realpower variables, linear programming with AC power flow variables.							
UNIT-IV (10 Hrs.)	Single area &Two areas Load Frequency Control: concept of single & two area Load frequency control:uncontrolled case and controlled case, tie-line bias control. Optimal two-area LF control-steadystate representation, performance Index and optimal parameter adjustment.							
UNIT-V (8 Hrs.)	Interchange Evaluation and Power Pools Economy Interchange, Economy interchangeEvaluation, Interchange Evaluation with unit commitment, Multiple Interchange contracts after the fact production costing, Transmission Losses in transaction Evaluation, other types ofInterchange, power pools.							
Text Books:								

1.	Power Generation, Operation and Control - by A.J.Wood and B.F.Wollenberg, 2 nd edition, John wiley& sons Inc. 1984.
2.	Power system operation and control - PSR Murthy, 1 st edition, B.S publication. 2008.
Reference Book:	
1.	Modern Power System Analysis - by I.J.Nagrath&D.P.Kothari, Tata McGraw-Hill Publishing Company ltd, 2nd edition.
2.	Electrical Energy Systems Theory - by O.I.Elgerd, Tata McGraw-Hill Publishing Company Ltd, 2nd edition.
3.	Reactive Power Control in Electric Systems - by TJE Miller, John Wiley & sons.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1102	PC	3	0	0	3	25	75	3 Hrs.
POWER SYSTEM DYNAMICS & STABILITY								
Course Objectives: The main objectives of this subject are:								
1	To remember the dynamic characteristics of power system equipment,							
2	To recognize dynamic performance of power systems							
3	To illustrate the system stability and controls.							
Course Outcomes								
S.No	Out come							Knowledge Level
1	Choose the fundamental dynamic behavior and controls of power systems to perform basic stability analysis.							K4
2	Comprehend concepts in modeling and simulating the dynamic phenomena of power systems							K3
3	Interpret results of system stability studies Analyze theory and practice of modeling main power system components, such as synchronous machines, excitation systems and governors.							K4
SYLLABUS								
UNIT-I (10 Hrs)		Basic Concepts: Power system stability states of operation and system security – system dynamics – system model analysis of steady State stability and transient stability –Modeling of Synchronous Machine: Synchronous machine – park’s Transformation of Flux Linkages, Stator Voltage equation and torque equation-analysis of steady state performance.						
UNIT-II (10 Hrs.)		Excitation System: Excitation system modeling-excitation systems block Diagram-Dynamics of a synchronous generator connected to infinite bus – system model Synchronous machine model-stator equations rotor equations – Synchronous machine model with field circuit – one equivalent damper winding on q axis (model 1.1).						
UNIT-III (10 Hrs.)		Analysis of Single Machine System: Small signal analysis with block diagram – Computation of Heffron –Phillips constants - Representation Characteristic equation and application of Routh Hurwitz criterion synchronizing and damping torque analysis-Stability by Eigen value Approach.						
UNIT-IV (10 Hrs.)		Application of Power System Stabilizers: Basic concepts in applying PSS – Control signals – Structure and tuning of PSS – Washout circuit – Dynamic compensator analysis of single machine infinite bus system with and without PSS.						
UNIT-V (10 Hrs.)		Digital Simulation of Transient Stability: Swing equation, machine equations – Direct method of solution – Solution Techniques: Modified Euler method – RungeKutta method – Concept of multi machine stability.						
Text Books:								
1.	K. R. PADIYAR,” Power system dynamics “- second edition, B.S. Publications, 2008.							

2.	P.M. Anderson and A. A. Fouad, “Power system control and stability”, first edition, IEEE Press, 1997.
3.	Power System Stability and control by PrabhaKundur, first edition, TMH Publications, 1994.
4.	Power System Stability by Kimbark Vol. III, Wiely.
Reference Books:	
1.	R. Ramanujam, “Power Systems Dynamics”- PHI Publications.
2.	Computer Applications to Power Systems–Glenn.W.Stagg& Ahmed. H.El.Abiad, TMH.
3.	Computer Applications to Power Systems – M.A.Pai, TMH.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1103	PE	3	0	0	3	25	75	3 Hrs.
ELECTRIC POWER DISTRIBUTION SYSTEM								
(ELECTIVE-I)								
Course Objectives: The main objectives of this subject are:								
1	To study the detail functioning of electric power distribution network.							
2	To evaluate the reliability of the electrical distribution system.							
3	To make the students to understand the distribution system expansion planning procedures							
4	Study Distribution Automation							
Course Outcomes								
S.No	Out come							Knowledge Level
1	Be proficient in control and protection schemes for distribution systems.							K2
2	Carry out planning and reliability analysis of electrical distribution systems.							K2
3	Understand Distribution automation and its application in practice							K2
SYLLABUS								
UNIT-I (16 Hrs)	Distribution of Power, Management, Power Loads, Load Forecasting Short-term & Long-term, Power System Loading, Technological Forecasting. Advantages of Distribution Management System (D.M.S.) Distribution Automation: Definition, Restoration / Reconfiguration of Distribution Network, Different Methods and Constraints, Power Factor Correction							
UNIT-II (8 Hrs.)	Distribution system - reliability analysis – reliability concepts – Markov model – distribution network reliability – reliability performance							
UNIT-III (8 Hrs.)	Distribution system expansion - planning – load characteristics – load forecasting – design concepts – optimal location of substation – design of radial lines – solution technique.							
UNIT-IV (8 Hrs.)	Calculation of Optimum Number of Switches, Capacitors, Optimum Switching Device Placement in Radial, Distribution Systems, Sectionalizing Switches – Types, Benefits, Bellman’s Optimality Principle, Remote Terminal Units, Energy efficiency in electrical distribution & Monitoring							
UNIT-V (8 Hrs.)	Maintenance of Automated Distribution Systems, Difficulties in Implementing Distribution Automation in Actual Practice, Urban/Rural Distribution, Energy Management, AI techniques applied to Distribution Automation							
Text Books:								
1.	A.S. Pabla, “ Electric Power Distribution”, Tata McGraw Hill Publishing Co. Ltd., Fourth Edition.							
2.	Tuvar Goner, ‘Electrical Power Distribution System Engineering’, McGraw hill, 2008.							
3.	M.K. Khedkar, G.M. Dhole, “A Text Book of Electrical Power Distribution Automation”, University Science Press, New Delhi.							
Reference Books:								
1.	Anthony J Panseni, “Electrical Distribution Engineering”, CRC Press							

2.	James Momoh, “Electric Power Distribution, automation, protection & control”, CRCPress.
3.	Electric Power Distribution System Engineering, TuranGonen, McGraw-Hill Inc., New Delhi, 1986.
4.	Control and Automation of Electrical Power Distribution Systems, James Northcote-Green and Robert Wilson, CRC Taylor & Francis, New York, 2007

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1104	PE	3	0	0	3	25	75	3 Hrs.
HVDC TRANSMISSION								
(ELECTIVE-I)								
Course Objectives: The main objectives of this subject are:								
1	To introduce the students with various merits, schemes, basic equipment, and operation of HVDC transmission systems.							
2	To familiarize the students with the control of HVDC transmission systems, the filters for harmonic elimination and the interaction between HVAC and HVDC systems.							
3	To expose the students to the circuit breakers, MTDC systems and various protection schemes for HVDC transmission.							
Course Outcomes								
S.No								Knowledge Level
1	Summarize various merits and schemes of HVDC transmission.							K2
2	Understand the basic HVDC transmission equipment and their operation.							K2
3	Apply the control of HVDC systems and to employ filters of given specifications.							K3
4	Demonstrate the interaction between HVAC and HVDC system, and to illustrate the circuit breakers and MTDC systems.							K3
5	Understand the various protection schemes of HVDC transmission.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Limitation of EHVAC Transmission, Advantages of HVDC Technical economical reliability aspects. HVDC Transmission: General considerations, Power Handling Capabilities of HVDC Lines, Basic Conversion principles, static converter configuration. Types of HVDC links- Apparatus and its purpose.							
UNIT-II (10 Hrs.)	Static Power Converters: 6-pulse bridge circuit and 12-pulse converters, converter station and Terminal equipment, commutation process, Rectifier and inverter operation, equivalent circuit for converter – special features of converter transformers. Performance comparison of diametrical connection with 6-pulse bridge circuit.							
UNIT-III (10 Hrs.)	Control of HVDC Converters and systems : Constant current, constant extinction angle and constant Ignition angle control. Individual phase control and equidistant firing angle control, DC power flow control. Factors responsible for generation of Harmonics, voltage and current harmonics, effect of variation of α and μ . Filters for harmonic elimination.							
UNIT-IV (10 Hrs.)	Interaction between HVAC and DC systems – Voltage interaction, Harmonic instability problems and DC power modulation. Development of DC circuit Breakers, Multi-terminal DC links and systems; series, parallel and series-parallel systems, their operation and control.							
UNIT-V (10 Hrs.)	Transient over voltages in HVDC systems : Over voltages due to disturbances on DC side, over voltages due to DC and AC side line faults. Converter faults and protection in HVDC Systems: Converter faults, over current protection - valve group and DC line protection, circuit breakers. Over voltage protection of converters, surge arresters.							

Text Books:	
1.	S. Kamakshai and V. Kamaraju, 'HVDC Transmission', First Edition, Tata McGraw Hill Education Pvt. Ltd., New Delhi.
2.	K. R. Padiyar, 'High Voltage Direct Current Transmission', Third Edition, New Age Publishers, New Delhi.
3.	E. W. Kimbark, 'Direct Current Transmission', Volume 1, First edition, Wiley Inter Science – New York.
Reference Books:	
1.	J. Arrillaga, 'High Voltage Direct Current Transmission', Second Edition, IET Publishers, UK.
2.	Vijay K. Sood, 'HVDC and FACTS controllers: Applications of static converters in power systems', First Edition, Kluwer Academic Press, New York.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1105	PE	3	0	0	3	25	75	3 Hrs.
EHVAC TRANSMISSION								
(ELECTIVE-I)								
Course Objectives: The main objectives of this subject are:								
1	To study the estimation of transmission line parameters and field effect on EHV and UHV AC lines.							
2	To acquire knowledge on corona, RI and audible noise in EHV and UHV lines.							
3	To study the concept of voltage control and compensation problems in EHV and UHV transmission systems.							
Course Outcomes								
S.No								Knowledge Level
1	Calculate the transmission line parameters.							K4
2	Calculate the field effects on EHV and UHV AC lines.							K4
3	Determine the corona, RI and audible noise in EHV and UHV lines.							K4
4	Analyze voltage control and compensation problems in EHV and UHV transmission systems.							K4
SYLLABUS								
UNIT-I (10 Hrs)	E.H.V.A.C. Transmission, line trends and preliminary aspects, standard transmission voltages – power handling capacities and line losses – mechanical aspects. Calculation of line resistance and inductance: resistance of conductors, temperature rise of conductor and current carrying capacity. Properties of bundled conductors and geometric mean radius of bundle, inductance of two conductor lines and multi conductor lines, Maxwell's coefficient matrix. Line capacitance calculation. capacitance of two conductor line, and capacitance of multi conductor lines, potential coefficients for bundled conductor lines, sequence inductances and capacitances and diagonalization.							
UNIT-II (10 Hrs.)	Calculation of electro static field of AC lines - Effect of high electrostatic field on biological organisms and human beings. Surface voltage Gradient on conductors, surface gradient on two conductor bundle and cosine law, maximum surface voltage gradient of bundle with more than 3 sub conductors, Mangolt formula.							
UNIT-III (10 Hrs.)	Corona : Corona in EHV lines – corona loss formulae – attenuation of traveling waves due to corona – Audio noise due to corona, its generation, characteristics and limits, measurement of audio noise.							
UNIT-IV (10 Hrs.)	Power Frequency voltage control : Problems at power frequency, generalized constants, No load voltage conditions and charging currents, voltage control using synchronous condenser, cascade connection of components : Shunt and series compensation, sub synchronous resonance in series – capacitor compensated lines							
UNIT-V (10 Hrs.)	Static reactive compensating systems: Introduction, SVC schemes, Harmonics injected into network by TCR, design of filters for suppressing harmonics injected into the system.							

Text Books:	
1.	Extra High Voltage AC Transmission Engineering – Rakesh Das Begamudre, Wiley Eastern Ltd., New Delhi – 1987.
Reference Books:	
1.	EHV Transmission line reference book – Edison Electric Institute (GEC) 1986.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1106	PE	3	0	0	3	25	75	3 Hrs.
ARITFICIAL INTELLIGENCE TECHNIQUES								
(ELECTIVE-II)								
Course Objectives: The main objectives of this subject are:								
1	To study the fuzzy logic expert system and know to design the fuzzy logic controllers.							
2	To study the concept of neural network and different types of neural networks training algorithms.							
3	To understand the concept of genetic algorithm and its application in optimization.							
4	To know the applications of AI Techniques in electrical engineering.							
Course Outcomes								
S.No								Knowledge Level
1	Analyze and design fuzzy logic systems.							K4
2	Understand the neural networks and analyze the different problems using training algorithms in neural networks.							K4
3	Develop algorithms using genetic algorithm for optimization.							K3
4	Apply AI Techniques in electrical engineering.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction:Approaches to intelligent control, Architecture for intelligent control, Symbolic reasoning system, Rule-based systems, the AI approach, Knowledge representation - Expert systems.							
UNIT-II (10 Hrs.)	Fuzzy Logic System: Introduction to classical sets - properties, operations and relations; Fuzzy sets, membership,Uncertainty, operations, properties, fuzzy relations, cardinalities, membership functions. Fuzzy Logic System Components-Fuzzification, Membership value assignment, development ofrule base and decision making system, defuzzification to crisp sets, defuzzification methods.							
UNIT-III (10 Hrs.)	Artificial Neural Networks:Humans and Computers, Concept of Artificial Neural Networks and its basic mathematical model, McCulloch-Pitts neuron model, simple perceptron, Adaline and Madaline, Feed-forward Multilayer Perceptron, Learning and Training the neural network using error back propagation training, Perceptron convergence theorem, Limitationsand applications of the Perceptron model, Generalized delta learning rule, Hopfield network, Self-organizing network and Recurrent network.							
UNIT-IV (10 Hrs.)	Genetic algorithms &Modelling Introduction-encoding-fitness function-reproductionoperators-genetic operators-cross over and mutation-generational cycle-convergence of geneticalgorithm							
UNIT-V (10 Hrs.)	Application of AI Techniques-load forecasting-load flow studies-economic loaddispatch-load frequency control-reactive power control-speed control of dc and ac motors							
Text Books:								
1.	Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications by Rajasekharan and Pai – PHI Publication.							
2.	J M Zurada , “An Introduction to ANN”, Jaico PublishingHouse							
3.	Simon Haykins, “Neural Networks”, Prentice Hall							

Reference Books:	
1.	Introduction to Artificial Neural Systems - Jacek M. Zuarda, Jaico Publishing House, 1997.
2.	Timothy Ross, "Fuzzy Logic with Engg.Applications", McGraw.Hill
3.	Driankov, Dimitra, "An Introduction to Fuzzy Control", NarosaPublication
4.	Golding, "Genetic Algorithms", Addison-Wesley PublishingCom

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1107	PE	3	0	0	3	25	75	3 Hrs.
OPTIMIZATION TECHNIQUES								
(ELECTIVE-II)								
Course Objectives: The main objectives of this subject are:								
1	To learn the essential optimization techniques for the application of engineering problems.							
2	To study the optimization techniques for linear and non-linear programming problems.							
Course Outcomes								
S.No	Out Come							Knowledge Level
1	Apply the optimization techniques to various engineering and other problems.							K4
SYLLABUS								
UNIT-I (8 Hrs)	Introduction to Optimization: Introduction, Historical Development, Engineering applications of Optimization, Statement of Optimization Problem.							
UNIT-II (10 Hrs.)	Classical Optimization Techniques: Introduction, Single variable optimization, Multivariable optimization with no constraints; Multivariable optimization with Equality constraints -Solution by Direct Substitution method, Method of Lagrangian multipliers; Multivariable optimization with inequality constraints: Kuhn-Tucker conditions.							
UNIT-III (10 Hrs.)	Linear Programming: Introduction, Applications of Linear Programming, Standard Form of a Linear Programming, Basic Terminology and Definitions, Exceptional cases, Simplex method, Big-M method,Two-phase method, Revised Simplex method, Duality, De generacy Principle.							
UNIT-IV (10 Hrs.)	Non-Linear Programming-I: Unconstrained One Dimensional Minimization Methods- Fibonacci Method, Quadratic Interpolation Method Non- Linear Programming: Unconstrained Optimnization: Univariate Method, Pattern Directions, Powell’s Method, Cauchy’s Method Or Steepest Descent Method, Powell’s Conjugate Direction Method.							
UNIT-V (10 Hrs.)	Non-Linear Programming-II: Constrained Optimization- Characteristics Of A Constrained Problem, Classification- Direct Methods, Indirect Methods- Interior Penalty Function Method, Exterior Penalty Function Method.							
Text Books:								
1.	Engineering Optimization: Theory and Applications 'By S.S. Rao, New Age International Publishers, revised Third Edition 2005.							
2.	Optimization for Engineering Design (Algorithms & Examples)“ by Kalyanmoy Deb, PHI Pvt. Ltd, New Delhi.							
Reference Books:								
1.	Optimization Techniques“ by Chander Mohan, Kusum Deep, New Age International Publishers, 2009.							
2.	Optimization Methods, Theory and Applications“ by HongleiXu, Song Wang , Soon-Yi Wu, Springer, 2015.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1108	PE	3	0	0	3	25	75	3 Hrs.
ADVANCED DIGITAL SIGNAL PROCESSING								
(ELECTIVE-II)								
Course Objectives: The main objectives of this subject are:								
1	To have knowledge on structures of different digital filters and Multirate DSP.							
2	To understand the linear prediction and optimum linear filters							
3	To analyze the signals with multi resolution properties using wavelet and KL transforms.							
4	To understand power spectrum estimation techniques in signal processing.							
Course Outcomes								
S.No	Out Come							Knowledge Level
1	Understand theory of different filters and algorithms							K2
2	Apply the multirate DSP in subband coding							K3
3	Estimate the optimal parameters of the signals based on the linear equations							K5
4	Apply wavelet and KL transforms for multiresolution analysis							K3
5	Estimate the power spectral parameters of given signals using different methods							K5
SYLLABUS								
UNIT-I (10 Hrs)	Overview of DSP, Digital filter design and structures: Basic FIR/IIR filter design & structures, design techniques of linear phase FIR filters, IIR filters by impulse invariance, bilinear transformation, FIR/IIR Cascaded lattice structures, and Parallel all pass realization of IIR.							
UNIT-II (10 Hrs.)	Multi rate DSP, Decimators and Interpolators, Sampling rate conversion, multistage decimator & interpolator, poly phase filters, QMF, digital filter banks, Applications in subband coding.							
UNIT-III (10 Hrs.)	Linear prediction & optimum linear filters, stationary random process, forward-backward linear prediction filters, solution of normal equations, AR Lattice and ARMA Lattice-Ladder Filters, Wiener Filters for Filtering and Prediction.							
UNIT-IV (10 Hrs.)	Introduction to multiresolution analysis and wavelets, wavelet properties, The Haar Wavelet, Wavelet decomposition and reconstruction, applications to denoising, KL Transform: Derivation, properties and applications.							
UNIT-V (10 Hrs.)	Estimation of Spectra from Finite-Duration Observations of Signals. Nonparametric Methods for Power Spectrum Estimation, Parametric Methods for Power Spectrum Estimation, Minimum Variance Spectral Estimation, Eigenanalysis Algorithms for Spectrum Estimation.							
Text Books:								
1.	Moon & Stirling, “Mathematical Methods and Algorithms for Signal Processing”, Prentice Hall, 2000.							
2.	P. P. Vaidyanathan, “Multirate systems and filter banks”, Prentice Hall, 2000.							

3.	Alan V. Oppenheim, Ronald W. Shafer “Digital Signal Processing”, 1st edition, First edition, 2015.
4.	J.G.Proakis and D.G.Manolakis“Digital signal processing: Principles, Algorithm and Applications”, 4th Edition, Prentice Hall, 2007.
Reference Books:	
1.	Theory and Applications of Digital Signal Processing-Lourens R. Rebinar&Bernold
2.	N. J. Fliege, “Multirate Digital Signal Processing: Multirate Systems -Filter Banks – Wavelets”, 1st Edition, John Wiley and Sons Ltd, 1999.
3.	Bruce W. Suter, “Multirate and Wavelet Signal Processing”, 1st Edition, Academic Press, 1997.
4.	M. H. Hayes, “Statistical Digital Signal Processing and Modeling”, John Wiley & Sons Inc., 2002.
5.	S.Haykin, “Adaptive Filter Theory”, 4th Edition, Prentice Hall, 2001.
6.	D.G.Manolakis, V.K. Ingle and S.M.Kogon, “Statistical and Adaptive Signal Processing”, McGraw Hill, 2000.
Web links	
1.	https://nptel.ac.in/courses/108108109/

Code	Category	L	T	P	C	I.M	E.M	Exam
M19RD1101	RD	2	0	0	2	25	75	3 Hrs.
RESEARCH METHODOLOGY AND IPR								
(Common to CST,CS,PSA, IT & CAD)								
Course Objectives:								
1.	To bring awareness on Research Methodology and research ethics.							
2.	Familiarize the concepts of IPR.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Analyze research related information							K4
2.	Formulate a Research Proposals and Publish papers with research ethics							K6
3.	Award for Intellectual Property Rights like Patents, Trade and Copyrights							K5
4.	Analyze Various Intellectual Property Rights							K4
5.	AssessNew Developments of IPRs in National and International level							K5
SYLLABUS								
UNIT-I (6 Hrs)	Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations							
UNIT-II (6 Hrs)	Effective literature studies approaches, analysis Plagiarism, Research ethics, Effective technical writing, how to write report, Paper Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee							
UNIT-III (6 Hrs)	Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.							
UNIT-IV (4 Hrs)	Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.							
UNIT-V (6 Hrs)	New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.							
Text Books:								
1.	Stuart Melville and Wayne Goddard, —Research methodology: an introduction for science & engineering students’l							
2.	Wayne Goddard and Stuart Melville, —Research Methodology: An Introductionll							
3.	Ranjit Kumar, 2nd Edition, —Research Methodology: A Step by Step Guide for beginnersl							
Reference Books:								
1.	Halbert, —Resisting Intellectual Propertyll, Taylor & Francis Ltd, 2007.							
2.	Mayall, —Industrial Designll, McGraw Hill, 1992.							
3.	Niebel, —Product Designll, McGraw Hill, 1974.							
4.	Asimov, —Introduction to Designll, Prentice Hall, 1962.							

5.	Robert P. Merges, Peter S. Menell, Mark A. Lemley, — Intellectual Property in New Technological Age, 2016.
6.	T. Ramappa, —Intellectual Property Rights Under WTO, S. Chand, 2008

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 PS 1109	PC	0	0	4	2	25	75	3 Hrs.
POWER SYSTEM SIMULATION LABORATORY-I								
Course Objectives:								
1.	To practice the basic theories of Electrical Power system.							
2.	To provide hands-on experience to the students, so that they are able to apply theoretical concepts in practice.							
3.	To use computer simulation tools such as MATLAB to carry out design experiments as it is a key analysis tool of engineering design.							
4.	To give a specific design problem to the students, which after completion they will verify using the simulation software or hardware implementation.							
5.	To understand the modeling of various aspects of Power System analysis and develop the MATLAB programming.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Verify the theoretical concepts of various aspects of Power System analysis.							K3
2.	Demonstrate the ability to identify, formulate and solve Power System engineering problems.							K4
3.	Design and conduct experiments, analyze and interpret data.							K5
4.	Design a electrical systems or process as per needs and specifications.							K5
5.	Use modern engineering tools, software's and equipment to analyze problem.							K5
List of Experiments								
1.	To write MATLAB programs for solving the given electrical networks.							
2.	To solve the given non-linear equation using Gauss Seidel and Newton-Raphson Techniques.							
3.	To find the Y-Bus of given power system by Direct Inspection method.							
4.	Obtain the power flow solution for the given system using Gauss-Seidel method.							
5.	Obtain the power flow solution for the given system using Newton-Raphson method.							
6.	Obtain the power flow solution for the given system using Fast Decoupled method.							
7.	Formation of Z-Bus by building Algorithm.							
8.	To verify the frequency and power deviations due to step-load changes in a two-area system using SIMULINK.							
9.	To obtain the swing curve for the given system and determine the stability using SIMULINK model.							
10.	Obtain the sequence components for the line currents using SIMULINK & MATLAB.							
11.	To design a state feedback controller for the load frequency control of an isolated power system using pole-placement technique and verify its effects.							
12.	To verify the effects of PID control on Automatic Voltage Regulator System of the generator by obtaining its terminal voltage step response.							
13.	To find the optimal dispatch and total cost of generation for three thermal power plants by neglecting transmission losses.							

Reference Books:	
1.	Elements of Power System Analysis" by William Stevenson, McGraw Hill Higher Education; 4th Revised edition.
2.	Power System Analysis HadiSaadat, PSA Publishing, 2010.
3.	Modern Power System Analysis" by D.P. Kothari and I.J. Nagrath, McGraw-Hill Companies, Incorporated, 2006.
4.	Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers" by RudraPratap, Oxford University Press, 2010
5.	MATLAB: An Introduction with Applications, 5th Edition Amos Gilat, Wiley Global Education, 2014.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 PS 1110	PC	0	0	4	2	25	75	3 Hrs.
POWER SYSTEM LABORATORY								
Course Objectives:								
1.	To obtain hands on experience for applying theoretical concepts of power system in practice.							
2.	To understand the modeling of power system components.							
3.	To practice the basic theories of power system protection systems.							
4.	To take up specific design problems on electro-mechanical and static relays.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Evaluate the theoretical concepts of power systems.							K4
2.	Demonstrate the modeling of key components of the power systems.							K3
3.	Understand the working of power system protection systems.							K2
4.	Compare the results of theoretical analyses with the practical ones.							K5
5.	Design protection systems of specified requirements.							K6
List of Experiments								
1.	Determination of positive, negative and zero sequence impedances of a 3 - phase transformer							
2.	Determination of positive, negative and zero sequence reactances of a synchronous machine through symmetrical components method							
3.	Determination of positive, negative and zero sequence reactances of a synchronous machine through fault analysis							
4.	Power angle characteristics of a salient pole synchronous machine							
5.	Determination of dielectric strength of insulating oil							
6.	Determination of equivalent circuit of a 3-winding transformer							
7.	Determination of ABCD parameters of transmission lines							
8.	Characteristics of electro-magnetic type over current relay							
9.	DMT and IDMT characteristics of a static overvoltage relay							
10.	DMT and IDMT characteristics of a static undervoltage relay							
11.	Protection of a transformer with differential relay							
12.	Characteristics of a negative sequence current relay							
Reference Books:								
1.	“Power System Protection and Switchgear" by Badri Ram and D. N. Vishwakarma, 2 nd Edition, Tata McGraw-Hill Education, 2017.							
2.	“Electrical Power Systems” by C. L. Wadhwa, 7 th Edition, New Age International, 2016.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0001	AC	2	0	0	0	0	0	--
ENGLISH FOR RESEARCH PAPER WRITING								
Course Objectives:								
1.	Understand how to improve your writing skills and level of readability							
2.	Learn about what to write in each section.							
3.	Understand the skills needed when writing a Title.							
4.	Ensure the good quality of paper at very first-time submission							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand that how to improve your writing skills and level of readability							K2
2.	Learn about what to write in each section							K2
3.	Understand the skills needed when writing a Title Ensure the good quality of paper at very first time submission							K2
SYLLABUS								
UNIT-I (4Hrs)	Planning and Preparation, Word Order, Breaking up long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness							
UNIT-II (4Hrs)	Clarifying Who Did What, Highlighting Your Findings, Hedging And Criticizing, Paraphrasing and Plagiarism, Sections of a Paper.							
UNIT-III (4Hrs)	Abstracts, Introduction, Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check.							
UNIT-IV (4Hrs)	Key skills are needed when writing a Title, key skills are needed when writing an abstract, key skills are needed when writing an introduction, skills needed when writing a Review of the Literature, skills are needed when writing the Methods.							
UNIT-V (4Hrs)	skills needed when writing the Results, skills are needed when writing the Discussion, skills are needed when writing the Conclusions, useful phrases, how to ensure paper is as good as it could possibly be the first- time submission							
Text Books:								
1.	Goldbort R (2006) Writing for Science, Yale University Press (available on Google Books)							
2.	Day R (2006) How to Write and Publish a Scientific Paper, Cambridge University Press							
Reference Books:								
1.	Highman N (1998), Handbook of Writing for the Mathematical Sciences, SIAM. Highman’s book.							
2.	Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0002	AC	2	0	0	0	0	0	--
DISASTER MANAGEMENT								
Course Objectives:								
1.	Learn to demonstrate a critical understanding of key concepts in disaster risk reduction and humanitarian response.							
2.	Critically evaluate disaster risk reduction and humanitarian response policy and practice from multiple perspectives.							
3.	Develop an understanding of standards of humanitarian response and practical relevance in specific types of disasters and conflict situations.							
4.	Critically understand the strengths and weaknesses of disaster management approaches, planning & programming in different countries, particularly their home country or the countries they work in.							
SYLLABUS								
UNIT-I (4Hrs)	Disaster: Definition, Factors And Significance; Difference Between Hazard And Disaster; Natural And Manmade Disasters: Difference, Nature, Types And Magnitude.							
UNIT-II (4Hrs)	Economic Damage, Loss Of Human And Animal Life, Destruction Of Ecosystem. Natural Disasters: Earthquakes, Volcanisms, Cyclones, Tsunamis, Floods, Droughts And Famines, Landslides And Avalanches, Man-made disaster: Nuclear Reactor Meltdown, Industrial Accidents, Oil Slicks And Spills, Outbreaks Of Disease And Epidemics, War And Conflicts							
UNIT-III (4Hrs)	Disaster Prone Areas In India: Study Of Seismic Zones; Areas Prone To Floods And Droughts, Landslides And Avalanches; Areas Prone To Cyclonic And Coastal Hazards With Special Reference To Tsunami; Post-Disaster Diseases And Epidemics.							
UNIT-IV (4Hrs)	Disaster Preparedness And Management Preparedness: Monitoring Of Phenomena Triggering A Disaster Or Hazard; Evaluation Of Risk: Application Of Remote Sensing, Data From Meteorological And Other Agencies, Media Reports: Governmental And Community Preparedness.							
UNIT-V (4Hrs)	Risk Assessment Disaster Risk: Concept And Elements, Disaster Risk Reduction, Global And National Disaster Risk Situation. Techniques Of Risk Assessment, Global Co-Operation In Risk Assessment And Warning, People’s Participation In Risk Assessment. Strategies for Survival. Disaster Mitigation Meaning, Concept And Strategies Of Disaster Mitigation, Emerging Trends In Mitigation. Structural Mitigation And Non-Structural Mitigation, Programs Of Disaster Mitigation In India.							
Text Books:								
1.	R. Nishith, Singh AK, Disaster Management in India: Perspectives, issues and strategies New Royal Book Company.							
2.	Sahni, PardeepEt.Al. (Eds.),Disaster Mitigation Experiences And Reflections, Prentice Hall Of India, New Delhi.							
3.	Goel S. L. Disaster Administration And Management Text And Case Studies ,Deep &Deep Publication Pvt. Ltd., New Delhi.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0003	AC	2	0	0	0	0	0	--
SANSKRIT FOR TECHNICAL KNOWLEDGE								
Course Objectives:								
1.	To get a working knowledge in illustrious Sanskrit, the scientific language in the world							
2.	Learning of Sanskrit to improve brain functioning							
3.	Learning of Sanskrit to develop the logic in mathematics, science & other subjects enhancing the memory power							
4.	The engineering scholars equipped with Sanskrit will be able to explore the huge knowledge from ancient literature							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understanding basic Sanskrit language.							K2
2	Ancient Sanskrit literature about science & technology can be understood.							K2
3	Being a logical language will help to develop logic in students.							K6
SYLLABUS								
UNIT-I (7Hrs)	Alphabets in Sanskrit, Past/Present/Future Tense, Simple Sentences							
UNIT-II (7Hrs)	Order, Introduction of roots, Technical information about Sanskrit Literature							
UNIT-III (7Hrs)	Technical concepts of Engineering-Electrical, Mechanical, Architecture, Mathematics							
Text Books:								
1.	“AbhyasaPustakam” – Dr.Vishwas, Samskrita-Bharati Publication, New Delhi							
2.	“Teach Yourself Sanskrit” PrathamaDiksha-VempatiKutumbasastry, Rashtriya Sanskrit Sansthanam, New Delhi Publication							
3.	“India’s Glorious Scientific Tradition” Suresh Soni, Ocean books (P) Ltd., New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0004	AC	2	0	0	0	0	0	--
VALUE EDUCATION								
Course Objectives:								
1.	Understand the value of education and self- development							
2.	Imbibe good values in students							
3.	Let should know about the importance of character							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Knowledge of self-development							K1
2	Learn the importance of Human values							K2
3	Developing the overall personality							K3
SYLLABUS								
UNIT-I (4Hrs)	Values and self-development –Social values and individual attitudes. Work ethics, Indian vision of humanism.Moral and non- moral valuation. Standards and principles. Value judgements.							
UNIT-II (4Hrs)	Importance of cultivation of values. Sense of duty. Devotion, Self-reliance. Confidence, Concentration. Truthfulness, Cleanliness. Honesty, Humanity. Power of faith, National Unity, Patriotism, Love for nature ,Discipline							
UNIT-III (4Hrs)	Personality and Behavior Development - Soul and Scientific attitude. Positive Thinking. Integrity and discipline. Punctuality, Love and Kindness. Avoid fault Thinking. Free from anger, Dignity of labour. Universal brotherhood and religious tolerance. True friendship. Happiness Vs suffering, love for truth. Aware of self-destructive habits. Association and Cooperation, Doing best for saving nature							
UNIT-IV (4Hrs)	Character and Competence –Holy books vs Blind faith.Self-management and Good health. Science of reincarnation. Equality, Nonviolence ,Humility, Role of Women. All religions and same message. Mind your Mind, Self-control. Honesty, Studying effectively.							
Text Books:								
1.	Chakroborty, S.K. “Values and Ethics for organizations Theory and practice”, Oxford University Press, New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0005	AC	2	0	0	0	0	0	--
CONSTITUTION OF INDIA								
Course Objectives:								
1.	Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective.							
2.	To address the growth of Indian opinion regarding modern Indian intellectuals’ constitutional role and entitlement to civil and economic rights as well as the emergence of nationhood in the early years of Indian nationalism							
3.	To address the role of socialism in India after the commencement of the Bolshevik Revolution in 1917 and its impact on the initial drafting of the Indian Constitution							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Discuss the growth of the demand for civil rights in India for the bulk of Indians before the arrival of Gandhi in Indian politics.							K2
2	Discuss the intellectual origins of the framework of argument that informed the conceptualization of social reforms leading to revolution in India.							K2
3	Discuss the circumstances surrounding the foundation of the Congress Socialist Party [CSP] under the leadership of Jawaharlal Nehru and the eventual failure of the proposal of direct elections through adult suffrage in the Indian Constitution.							K2
4	Discuss the passage of the Hindu Code Bill of 1956.							K2
SYLLABUS								
UNIT-I (4Hrs)	History of Making of the Indian Constitution: History , Drafting Committee, (Composition & Working)							
UNIT-II (4Hrs)	Philosophy of the Indian Constitution: Preamble ,Salient Features							
UNIT-III (4Hrs)	Fundamental Rights,Right to Equality, Right to Freedom,Right against Exploitation, Right to Freedom of Religion,Cultural and Educational Rights,Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties							
UNIT-IV (4Hrs)	Organs of Governance: Parliament, Composition, Qualifications and Disqualifications ,Powers and Functions, Executive, President , Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions Local Administration: District’s Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative, CEO of Municipal Corporation, Pachayati raj: Introduction, PRI: Zilla Panchayat. Elected officials and their roles, CEO Zilla Panchayat: Position and role. Block level: Organizational Hierarchy (Different departments),Village level: Role of Elected and Appointed officials, Importance of grass root democracy							
UNIT-V (4Hrs)	Election Commission: Election Commission: Role and Functioning. Chief Election Commissioner and Election Commissioners. State Election Commission: Role and Functioning. Institute and Bodies for the welfare of SC/ST/OBC and women.							

Text Books:	
1.	The Constitution of India, 1950 (Bare Act), Government Publication.
2.	Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.
3.	M. P. Jain, Indian Constitution Law, 7th Edn., LexisNexis, 2014.
4.	D.D. Basu, Introduction to the Constitution of India, LexisNexis, 2015.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0006	AC	2	0	0	0	0	0	--
PEDAGOGY STUDIES								
SYLLABUS								
UNIT-I (4Hrs)	Introduction and Methodology: Aims and rationale, Policy background, Conceptual framework and terminology Theories of learning, Curriculum, Teacher education. Conceptual framework, Research questions. Overview of methodology and Searching.							
UNIT-II (4Hrs)	Thematic overview: Pedagogical practices are being used by teachers in formal and informal classrooms in developing countries, Curriculum, Teacher education.							
UNIT-III (4Hrs)	Evidence on the effectiveness of pedagogical practices, Methodology for the in depth stage: quality assessment of included studies. How can teacher education (curriculum and practicum) and the school curriculum and guidance materials best support effective pedagogy.							
UNIT-IV (4Hrs)	Theory of change, Strength and nature of the body of evidence for effective pedagogical practices Pedagogic theory and pedagogical approaches. Teachers’ attitudes and beliefs and Pedagogic strategies. Professional development: alignment with classroom practices and follow-up support							
UNIT-V (4Hrs)	Research gaps and future directions: Research design, Contexts, Pedagogy, Teacher education, Curriculum and assessment, Dissemination and research impact.							
Text Books:								
1.	Ackers J, Hardman F (2001) Classroom interaction in Kenyan primary schools, Compare, 31 (2): 245-261.							
2.	Agrawal M (2004) Curricular reform in schools: The importance of evaluation, Journal of Curriculum Studies, 36 (3): 361-379.							
3.	Akyeampong K (2003) Teacher training in Ghana - does it count? Multi-site teacher education research project (MUSTER) country report 1. London: DFID.							
4.	Akyeampong K, Lussier K, Pryor J, Westbrook J (2013) Improving teaching and learning of basic maths and reading in Africa: Does teacher preparation count? International Journal of Educational Development, 33 (3): 272–282.							
5.	Alexander RJ (2001) Culture and pedagogy: International comparisons in primary education. Oxford and Boston: Blackwell.							
6.	Chavan M (2003) Read India: A mass scale, rapid, „learning to read“ campaign.							
7.	www.pratham.org/images/resource%20working%20paper%202.pdf.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0007	AC	2	0	0	0	0	0	--
STRESS MANAGEMENT BY YOGA								
Course Objectives:								
1.	To achieve overall health of body and mind							
2.	To overcome stress							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Develop a healthy mind in a healthy body thus improving social health also.							K2
2	Improve efficiency							K2
SYLLABUS								
UNIT-I (7Hrs)	Definitions of Eight parts of yoga (Ashtanga)							
UNIT-II (7Hrs)	Yam and Niyam. Do's and Don'ts in life. i) Ahinsa, satya, astheya, brahmacharya and aparigraha ii) Shaucha, santosh, tapa, swadhyay, ishvarapranidhana							
UNIT-III (7Hrs)	Asan and Pranayam i) Various yog poses and their benefits for mind & body ii)Regularization of breathing techniques and its effects-Types pranayama							
Text Books:								
1.	‘Yogic Asanas for Group Training-Part-I’ : Janardan Swami YogabhyasiMandal, Nagpur							
2.	“Raja Yoga or conquering the Internal Nature” by Swami Vivekananda, AdvaitaAshrama (Publication Department), Kolkata							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0008	AC	2	0	0	0	0	0	--
PERSONALITY DEVELOPMENT THROUGH LIFEENLIGHTENMENT SKILLS								
Course Objectives:								
1.	To learn to achieve the highest goal happily							
2.	To become a person with a stable mind, pleasing personality and determination							
3.	To awaken wisdom in students							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Study of Shrimad-Bhagwad-Geeta will help the student in developing his personality and achieve the highest goal in life.							K2
2	The person who has studied Geeta will lead the nation and mankind to peace and prosperity.							K2
3	Study of Neetishatakam will help in developing versatile personality of students.							K2
SYLLABUS								
UNIT-I (7Hrs)	Neetisatakam-Holistic development of personality Verses- 19,20,21,22 (wisdom), Verses- 29,31,32 (pride & heroism) Verses- 26,28,63,65 (virtue),Verses- 52,53,59 (don'ts),Verses- 71,73,75,78 (do's)							
UNIT-II (7Hrs)	Approach to day to day work and duties. ShrimadBhagwadGeeta : Chapter 2-Verses: 41, 47,48 Chapter 3-Verses 13, 21, 27, 35, Chapter 6-Verses 5,13,17, 23, 35, Chapter 18-Verses 45, 46, 48.							
UNIT-III (7Hrs)	Statements of basic knowledge, ShrimadBhagwadGeeta: Chapter2-Verses 56, 62, 68, Chapter 12 -Verses 13, 14, 15, 16,17, 18, Personality of Role model. ShrimadBhagwadGeeta: Chapter2-Verses 17, Chapter 3-Verses 36,37,42, Chapter 4-Verses 18, 38,39, Chapter18 – Verses 37,38,63							
Text Books:								
1.	‘Srimad Bhagavad Gita” by Swami SwarupanandaAdvaita Ashram (Publication Department), Kolkata							
2.	Bhartrihari’s Three Satakam (Niti-sringar-vairagya) by P.Gopinath,Rashtriya Sanskrit Sansthanam, New Delhi.							



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

M.TECH (POWER SYSTEM & AUTOMATION)

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

(With effect from 2019-2020 Admitted Batch onwards)

II-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Int. Marks	External Marks	Total Marks
M19 PS1201	Real Time Control of Power Systems	PC	3	3	0	0	25	75	100
M19 PS 1202	Advanced Power Electronic Converters	PC	3	3	0	0	25	75	100
#PE-III	Program Elective-III	PE	3	3	0	0	25	75	100
#PE-IV	Program Elective-IV	PE	3	3	0	0	25	75	100
M19 PS 1209	Power System Simulation Laboratory-II	PC	2	0	0	4	25	75	100
M19 PS 1210	Power Electronics for Renewable Integration Laboratory	PC	2	0	0	4	25	75	100
M19 PS 1211	Mini Project with Seminar	MP	2	0	0	4	100	--	100
#AC-2	Audit Course-2*	AC	0	2	0	0	0	0	--
Total			18	14	0	12	250	450	700

	Code	Course
#PE-III	M19 PS1203	Power Quality Issues and Remedies
	M19 PS1204	Digital Protection of Power Systems
	M19 PS1205	Power System Transients
#PE-IV	M19 PS1206	Smart Grids Technologies
	M19 PS1207	Renewable Energy Systems
	M19 PS1208	Advanced Control Theory

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1201	PC	3	0	0	3	25	75	3 Hrs.
REAL TIME CONTROL OF POWER SYSTEMS								
Course Objectives:								
1.	To understand the importance of state estimation in power systems.							
2.	To know the importance of security and contingency analysis.							
3.	To learn about SCADA, its objectives and its importance in power systems.							
4.	To know the significance of voltage stability analysis.							
5.	To know about the PMU and its placement in power systems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand Supervisory control and data acquisition and EMS.							K2
2.	Analyze the voltage stability of the power system network.							K4
3.	Apply knowledge in placement of PMU for power system network							K3
4.	Solve the state estimation, and contingency in power system.							K4
SYLLABUS								
UNIT-I (10 Hrs)	State Estimation: Different types of State Estimations, Formulation of state Estimation problem, Various criterion's of state estimation, Theory of WLS state estimation, Networkobservability, Pseudo measurements, Bad data observability, Bad data detection, identification and elimination							
UNIT-II (10 Hrs)	Security and Contingency Evaluation : Security concept, Security Analysis and monitoring, Types of contingency, Contingency Analysis of power system network by using AC Power flow and DC power flow methods, Contingency selection, Network sensitivity factors and their calculation, Concentric relaxation, Bounding.							
UNIT-III (10 Hrs)	Computer Control of Power Systems : Need for real time and computer control of power systems, operating states of a power system-state transition diagram, SCADA - functions of Supervisory control and Data Acquisition systems and implementation considerations, Energy Control Centers (ECS/EMS),software requirements for implementing the above functions.							
UNIT-IV (10 Hrs)	Voltage Stability: Voltage collapse, Voltage security, Relation of voltage stability to rotor angle stability. Types of voltage stability, Static Voltage stability analysis, `P-V` curves and `Q-V` curves, Voltage stability static indices, Improvement of voltage stability, Reactive power optimization.							
UNIT-V (8 Hrs)	PMU:Wide Area Monitoring Systems (WAMS), Placement of Phasor Measurement Units (PMUs), Phasor and Frequency Estimation, Enhanced State Estimation, observability analysis, Voltage Stability assessment and fault detection using Phasor Measurement.							
Text Books:								
1.	John J.Grainger and William D.Stevenson, Jr. : Power System Analysis, McGraw-Hill, 1994, International Edition.							
2.	Allen J.Wood and Bruce F.Wollenberg : Power Generation operation and control, John Wiley & Sons, 1984.							
3.	Phadke A G, Thorp J S, “Synchronized Phasor Measurements and Their Applications”, Springer, 2008							

Reference Books:	
1.	R.N.Dhar : Computer Aided Power Systems Operation and Analysis, Tata McGraw Hill, 1982.
2.	L.P.Singh : Advanced Power System Analysis and Dynamics, Wiley Eastern Ltd. 1986.
3.	PrabhaKundur : Power System Stability and Control -, McGraw Hill, 1994

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1202	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED POWER ELECTRONIC CONVERTERS								
Course Objectives:								
1.	Understand the operation of advanced power electronic circuit topologies.							
2.	Understand the PWM control strategiesinvolved.							
3.	Understand the operation of multilevel Inverter.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Analyze and design of AC-AC Converters for various types of loads.							K4
2.	Evaluate the power factor correction methods in AC-DC Converters.							K4
3.	Acquire knowledge about analysis and design of SMPS, PWM techniques.							K3
4.	Summarize the features of different multilevel inverter.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Single Phase AC Voltage Controllers-Introduction –Principle of ON-OFF control, Principle of phase angle control, Single phase Bidirectional controllers with Resistive Loads, Single phase controllers with Inductive loads, Three Phase AC Voltage controllers-Analysis of Controllers with star and delta connected resistive, resistive–inductive loads, single phase transformer tap changers -Effects of source and load inductances- numerical problems.							
UNIT-II (10 Hrs)	AC-DC converters- Introduction -Single phase and three phase dual Converters, Power factor improvements: Extinction angle control-symmetrical angle control - single phase sinusoidal PWM-Single phase series converters- Effects of source and load inductances - numerical problems							
UNIT-III (10 Hrs)	Switch-Mode Converter Power Supplies- high-frequency switching- push–pull converter, single-switch forward converter, single-switch fly-back converter, Comparison of Continuous and Discontinuous Conduction Performance, numerical problems.							
UNIT-IV (10 Hrs)	Voltage control of single-phase inverters - sinusoidal PWM –modified PWM – phase displacement Control, PWM Techniques- SHE PWM, Space Vector PWM (SVM), Hysteresis band current control PWM, Advanced modulation techniques-Trapezoidal, staircase, stepped, harmonic injection and delta modulation.							
UNIT-V (8 Hrs)	Multi-level inverters, Types of Multilevel Inverters- Diode-Clamped Multilevel Inverter, Features of Diode-Clamped Inverter, Improved Diode-Clamped Inverter- Flying-Capacitors Multilevel Inverter-Features of Flying-Capacitors Inverter- Cascaded Multilevel Inverter-Principle of Operation- Features of Cascaded Inverter- Comparisons of Multilevel Converters.							
Text Books:								
1.	Power Electronics-Md.H.Rashid –Pearson Education Third Edition- First Indian Reprint- 2008.							
2.	Power Converter Circuits – William Shepherd & Li Zhang-Yes Dee Publishing Pvt Ltd.							
3.	Pulse Width Modulation: Analysis and Performance in Multilevel Inverters, Satish Kumar Peddapelli.First Edition,2017 Walter de Gruyter GmbH, Berlin/Boston.							
4.	Modern power Electronics and AC Drives – B.K.Bose.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1203	PE	3	0	0	3	25	75	3 Hrs.
POWER QUALITY ISSUES AND MITIGATION								
Course Objectives:								
1.	To understand significance of power quality and power quality parameters.							
2.	To know types of transient over voltages and protection of transient voltages and devices for voltage regulation							
3.	To understand harmonics, their effects, harmonic indices and harmonic minimization techniques.							
4.	To know power quality aspects in distributed generation.							
5.	To understand various mitigation techniques for power quality issues in distribution system							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Acquire the knowledge on causes of power quality, power quality parameters.							K2
2.	Understand sources of transient over voltages and providing protection to transient over voltages and study devices for voltage regulation							K2
3.	Understand effects and sources of harmonics, and design filter for controlling harmonic distortion.							K4
4.	Describe power quality aspects in distributed generation and develop solutions to wiring and grounding problems.							K3
5.	Analyze the effect of various power quality issues in distribution system and their mitigation principles.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction Overview of Power Quality - Concern about the Power Quality - General Classes of PowerQuality Problems – Transients -Long-Duration Voltage Variations - Short-Duration VoltageVariations - Voltage Unbalance - Waveform Distortion - Voltage fluctuation - Power FrequencyVariations - Power Quality Terms - Voltage Sags and Interruptions - Sources of Sags andInterruptions – Nonlinear loads.							
UNIT-II (10 Hrs)	Transient Over Voltages Source of Transient Over Voltages - Principles of Over Voltage Protection - Devices for OverVoltage Protection - Utility Capacitor Switching Transients - Utility Lightning Protection – LoadSwitching Transient Problems Long Duration Voltage Variations Principles of Regulating the Voltage - Device for Voltage Regulation - Utility Voltage RegulatorApplication - Capacitor for Voltage Regulation - End-user Capacitor Application							
UNIT-III (10 Hrs)	Harmonic Distortion and solutions Voltage vs. Current Distortion - Harmonics vs. Transients - Power System Quantities underNon-sinusoidal Conditions - Harmonic Indices – Sources of harmonics - Locating Sources ofHarmonics – System Response Characteristics - Effects of Harmonic Distortion – Interharmonics- Harmonic Solutions Harmonic Distortion Evaluation - Devices for Controlling HarmonicDistortion - Harmonic Filter Design - Standards on Harmonics							
UNIT-IV (10 Hrs)	Distributed Generation and Power Quality Resurgence of Distributed Generation - DG Technologies - Interface to the Utility System Power Quality Issues - Operating Conflicts - Wiring and Grounding - Typical Wiring and Grounding Problems -Solution to Wiring and grounding Problems							

UNIT-V (8 Hrs)	Power Quality Enhancement Using Custom Power Devices Introduction to Custom Power Devices - Static VarCompensation for Distribution Systems – Static Series Compensator - Static Shunt Compensator(DSTATCOM)-Application of Custom Power Devices in Power Systems--P-Q theory – Control of P and Q – Dynamic Voltage Restorer (DVR) – Operation and control – Operation and control – Unified Power QualityConditioner (UPQC) – Operation and control.
Text Books:	
1.	Electrical Power Systems Quality, Dugan R C, McGranaghan M F, Santoso S, and Beaty H W, Second Edition, McGraw-Hill, 2002.
2.	Bhim Singh, Ambrish Chandra, Kamal Al-Haddad, “Power Quality Problems and Mitigation Techniques” Wiley Publications, 2015.
Reference Books:	
1.	Understanding Power Quality Problems: Voltage Sags and Interruptions, Bollen M H J, First Edition, IEEE Press; 2000.
2.	Power System Harmonics, Arrillaga J and Watson N R, Second Edition, John Wiley & Sons, 2003.
3.	Power Quality Enhancement Using Custom Power Devices – Power Electronics and Power Systems, Gerard Ledwich, Arindam Ghosh, Kluwer Academic Publishers, 1st ed,2002.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1204	PE	3	0	0	3	25	75	3 Hrs.
DIGITAL PROTECTION OF POWER SYSTEMS								
Course Objectives:								
1.	Study of Staticrelays							
2.	Study of Pilot relaying and Carrier current protection schemes.							
3.	Developing mathematical basis for protective relaying algorithms,							
4.	Study of Digitalprotection for transmission lines.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand the various types of comparators and their realization using static circuits.							K2
2.	Understand the realization of over current, distance and differential relays using comparators.							K2
3.	Estimate the current and voltage magnitudes from the sampled measurements.							K4
4.	Realize the various dynamic characteristics of digital relays for protection of transmission lines, transformers.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Basic Block diagram – Advantages of Static Relays – Comparators – Phase and amplitude Comparators – General Equations of Comparators – Analysis of Amplitude and Phase Comparators – Operating principles–Static Overcurrent relays – Differential relays – distance relays.							
UNIT-II (10 Hrs)	Pilot relaying and Carrier current protection schemes – Protection of Transmission lines – 3–zone protection schemes – carrier aided distance schemes – switched distance schemes – Transformer protection – mal operation of relays – Harmonic Restraint relay – Wavelet applications in transformer protection – Multi Input Comparator circuits – realization of Elliptical and Quadrilateral characteristics							
UNIT-III (10 Hrs)	Digital Protection: Developments in computer relaying – mathematical basis for protective relaying algorithms, Fourier Transforms – Discrete Fourier transforms – Wavelet transforms – Kalman filtering.							
UNIT-IV (10 Hrs)	Digital protection of transformers – Fourier based algorithms – finite duration impulse response filter based algorithms – least square curve fitting based algorithms – flux restrained current differential relay.							
UNIT-V (8 Hrs)	Digital protection of transmission lines – current based differential schemes – composite voltage and current based schemes – New developments in relaying principles – fundamentals of travelling wave protection – principle of travelling wave distance relay – adaptive relaying – fault location algorithms.							
Text Books:								
1.	Power System Protection – Static relays T.S.MadhavaRao, TMH, 2010.							
2.	Digital Protection for Power Systems A.T.Johns and S.K.Salman, 1995.							
3.	Computer relaying for power Systems A.G.Phake, James S.Thorp, John–Wiley and sons.							
Reference Books:								
1.	Protective relaying principles and applications J.Lewis Blackburn, Marcel & Dekker.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1205	PE	3	0	0	3	25	75	3 Hrs.
POWER SYSTEM TRANSIENTS								
Course Objectives:								
1.	To study the generation of switching transients and their control using circuit – theoretical concept.							
2.	To study the mechanism of lighting strokes and the production of lighting surges.							
3.	To study the propagation, reflection and refraction of travelling waves.							
4.	To study the impact of voltage transients caused by faults, circuit breaker action, load rejection on integrated power system.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Ability to understand and analyze power system operation, stability, control and protection.							K4
2.	To limit the effects of lightning over voltages in power systems							K3
3.	Understand the concepts of travelling wave theory for transient signals							K2
4.	Understand the various transient over voltages and their effects on integrated power system.							K2
SYLLABUS								
UNIT-I (10 Hrs)	INTRODUCTION AND SURVEY: Review and importance of the study of transients - causes for transients. RL circuit transient with sine wave excitation - double frequency transients - basic transforms of the RLC circuit transients. Different types of power system transients - effect of transients on power systems – role of the study of transients in system planning.							
UNIT-II (10 Hrs)	SWITCHING TRANSIENTS: Over voltages due to switching transients - resistance switching and the equivalent circuit for interrupting the resistor current - load switching and equivalent circuit - waveforms for transient voltage across the load and the switch - normal and abnormal switching transients. Current suppression - current chopping - effective equivalent circuit. Capacitance switching - effect of source regulation - capacitance switching with a restrike, with multiple restrikes. Illustration for multiple restriking transients - ferro resonance.							
UNIT-III (10 Hrs)	LIGHTNING TRANSIENTS: Review of the theories in the formation of clouds and charge formation - rate of charging of thunder clouds – mechanism of lightning discharges and characteristics of lightning strokes – model for lightning stroke - factors contributing to good line design - protection using ground wires - tower footing resistance - Interaction between lightning and power system.							
UNIT-IV (10 Hrs)	TRAVELING WAVES ON TRANSMISSION LINE COMPUTATION OF TRANSIENTS: Computation of transients - transient response of systems with series and shunt lumped parameters and distributed lines. Traveling wave concept - step response - Bewely’s lattice diagram - standing waves and natural frequencies - reflection and refraction of travelling waves.							
UNIT-V (8 Hrs)	TRANSIENTS IN INTEGRATED POWER SYSTEM: The short line and kilometric fault - distribution of voltages in a power system - Line dropping and load rejection - voltage transients on closing and reclosing lines - over voltage							

	induced by faults -switching surges on integrated system Qualitative application of EMTP for transient computation.
Text Books:	
1.	Allan Greenwood, 'Electrical Transients in Power Systems', Wiley Inter Science, New York, 2nd Edition, 1991.
2.	Pritindra Chowdhari, "Electromagnetic transients in Power System", John Wiley and Sons Inc., Second Edition, 2009.
3.	C.S. Indulkar, D.P.Kothari, K. Ramalingam, 'Power System Transients – A statistical approach', PHI Learning Private Limited, Second Edition, 2010.
Reference Books:	
1.	M.S.Naidu and V.Kamaraju, 'High Voltage Engineering', Tata McGraw Hill, Fifth Edition, 2013.
2.	R.D. Begamudre, 'Extra High Voltage AC Transmission Engineering', Wiley Eastern Limited, 1986.
3.	Y.Hase, Handbook of Power System Engineering," Wiley India, 2012.
4.	J.L.Kirtley, "Electric Power Principles, Sources, Conversion, Distribution and use," Wiley, 2012

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1206	PE	3	0	0	3	25	75	3 Hrs.
SMART GRID TECHNOLOGIES								
Course Objectives:								
1.	To understand various aspects of smart grid							
2.	To study various smart transmission and distribution technologies							
3.	To appreciate distribution generation and smart consumption							
4.	To know the regulations and market models for smart grid							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand technologies for smart grid							K2
2.	Appreciate the smart transmission as well distribution systems							K2
3.	Realize the distribution generation and smart consumption							K2
4.	Know the regulations and market models for smart grid							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Smart Grid: Definition, justification for smart grids, smart grid conceptual model, smart grid architectures, Interoperability, communication technologies, role of smart grids standards, intelligrid initiative, national smart grid mission (NSGM) by Govt. of India							
UNIT-II (10 Hrs)	Smart Transmission Technologies:Substation automation, Supervisory control and data acquisition (SCADA), energy management system (EMS), phasor measurement units (PMU), Wide area measurement systems (WAMS)							
UNIT-III (10 Hrs)	Smart Distribution Technologies: Distribution automation, outage management systems, automated meter reading (AMR), automated metering infrastructure (AMI), fault location isolation and service restoration (FLISR), Outage Management Systems (OMS), Energy Storage, Renewable Integration.							
UNIT-IV (10 Hrs)	Distributed Generation and Smart Consumption:Distributed energy resources (DERs), smart appliances, low voltage DC (LVDC) distribution in homes / buildings, home energy management system (HEMS), Net Metering, Building to Grid B2G, Vehicle to Grid V2G, Solar to Grid, Microgrid							
UNIT-V (8 Hrs)	Regulations and Market Models for Smart Grid:Demand Response, Tariff Design, Time of the day pricing (TOD), Time of use pricing (TOU), Consumer privacy and data protection, consumer engagement etc. Cost benefit analysis of smart grid projects.							
Text Books:								
1.	Clark W Gellings, “The Smart Grid, Enabling Energy Efficiency and Demand Side Response”- CRC Press, 2009.							
2.	Jean Claude Sabonnadière, NouredineHadjsaïd, “Smart Grids”, Wiley-ISTE, IEEE Press, May 2012							
Reference Books:								
1.	JanakaEkanayake, KithsiriLiyanage, Jianzhong. Wu, Akihiko Yokoyama, Nick Jenkins, “Smart Grid: Technology and Applications”- Wiley, 2012.							
2.	James Momoh, “Smart Grid: Fundamentals of Design and Analysis” – Wiley, IEEE Press, 2012.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1207	PE	3	0	0	3	25	75	3 Hrs.
RENEWABLE ENERGY SYSTEMS								
Course Objectives:								
1.	To understand the principles of renewable energy systems.							
2.	To Understand the Power Electronics Interface with Renewable energy system							
3.	To comprehensive understanding of the various renewable energy systems							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand world energy scenario and different Renewable energy systems.							K2
2.	Understand the different technologies available in Renewable energy systems.							K2
3.	Understand the need for Renewable energy technologies.							K2
4.	Get knowledge on New and upcoming renewable energy tehnologies							K2
SYLLABUS								
UNIT-I (10 Hrs)	Effects of Conventiinal Energy methods: Reserves of energy resources – Environmental aspects of conventional power plants- Impact of energy on economy – Renewable energy scenario in India – Potentials – Achievements.							
UNIT-II (10 Hrs)	Solar Energy: Systems: Solar thermal – Flat plate and concentrating collectors – Solar Photo Voltaic Conversion System – Working and design of Solar Cells and Panels – PV applications- MPPT methods- New avenues for growth of solar like floating Solar and Solar trees- Power electronic interface with Solar PV systems.							
UNIT-III (10 Hrs)	Wind Energy Systems:Basics of Wind Energy conversion system -Types of wind energy systems – Performance – Energy storage- Applications of Wind energy- Inter connected System – Safety and Environmental Aspects- modern trends in wind turbine technology.							
UNIT-IV (10 Hrs)	Biomass – Cogeneration– Biomass applications - Tidal energy – Wave energy – Mini and Micro hydro – Geothermal energy – Fuel cell systems - Modern trends in Renewable technology.							
UNIT-V (8 Hrs)	Distributed Generation in power systems: Distributed Generation, Voltage effects, Thermal Limits, Impact of Embedded Generation on the power System Power Quality Disturbances, Islanding- Economics of distributed Genration.							
Text Books:								
1.	Renewable Energy in Power Systems, by Leon Ferris, David Infield, Wiley Publishers, July 2008.							
2.	Non conventional Energy Sources. By G.D Rai, Khanna Publishers, New Delhi							
3.	Renewable Energy Resources” by John Twidell and Tony Weir, Taylor & Francis, Second edition, 2006.							
Reference Books:								
1.	Solar Energy-Fundamental Design, modelling and application by G N Tiwary: Narosa Publishing house, New Delhi							
2.	Renewable energy, Power for a sustainable future. By Godfrey Boyle: Oxford University press U.K							
3.	Renewable Energy Systems by Henrik Lund, Elsevier Inc. 2014.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1208	PE	3	0	0	3	25	75	3 Hrs.
ADVANCED CONTROL THEORY								
Course Objectives:								
1.	To facilitate the evolution of state variable approach for the analysis of control systems.							
2.	To examine the importance of controllability and observability in modern control engineering.							
3.	To enable students to analyze various types of nonlinearities & construction of trajectories using describing functions and phase plane analysis.							
4.	To study the analysis of stability and instability of continuous time invariant system							
5.	To facilitate the evolution of state variable approach for the analysis of control systems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understanding the state variable approach is suitable for higher order.							K2
2.	Analyze the concepts of controllability and observability.							K4
3.	Analyze the various non-linearities through describing functions and phase plane analysis.							K4
4.	Analyze typical issues of stability and instability of continuous time invariant systems.							K4
SYLLABUS								
UNIT-I (10 Hrs)	State Space Analysis & Techniques : The Concept of State and State Models, State Diagram, State Space and State Trajectory, State Space Representation using Phase Variable and Canonical Variables, Solution of State Equation, State Transition Matrix and its Properties, Eigen Values, Eigen Vectors, Model Matrix, Diagonalization, Generalized Eigen vectors, Computation of State Transition Matrix using Laplace Transformation, Power Series Method, Cayley-Hamilton Method, Similarity Transformation Method. Controllability and Observability Tests: Kalman's test, Gilbert's Test, Controllability and Observability Canonical Forms.							
UNIT-II (10 Hrs)	Pole Placement Techniques : Controller Design by State Feedback, Necessary and Sufficient Condition for Arbitrary Pole Placement-State Regulator Problem and State Regulator Design, Evaluation of State Feedback Gain Matrix K, Selection of Location of Desired Closed Loop Poles, State Observer Design, Full Order/Reduced Order Observer Design, Observer Based State Feedback Control, Separation Principle.							
UNIT-III (10 Hrs)	Non-Linear Control System: Introduction, Properties of Nonlinear System, Behavior of Non-Linear System, Classification of Nonlinearities, Common Physical Nonlinearities: Saturation, Friction, Backlash, Dead-Zone, Relay, On-Off Nonlinearity, Nonlinear Spring, Limit cycle, Jump resonance. Phase-Plane Method, Singular points, Stability of Nonlinear System, Construction of Phase Trajectories, Describing Functions Method, Stability Analysis by Describing Function Method. Lyapunov’s Stability Analysis, Stability Criterion, Direct Method of Lyapunov and the Linear Systems, Method of Construction of Lyapunov Functions for Nonlinear Systems.							
UNIT-IV (10 Hrs)	Optimal Control:Introduction to Optimal Control, Parameter Optimization: Servomechanism, Optimal Control Problem: Transfer Function and State Variable Approach, State Regulator Problem, Infinite Time Regulator Problem, Output Regulator and the Tracking Problem, Parameter Optimization: Regulators.							
UNIT-V (8 Hrs)	Digital Control Systems: Introduction to Discrete Time Systems, Necessary for Digital Control System, Spectrum Analysis ofSampling Process, Signal Reconstruction, Difference Equations, Z transforms, and the Inverse Ztransform, Pulse Transfer Function, Time Response of Sampled Data Systems, Stability using JuryCriterion, Bilinear Transformation.							

Text Books:	
1.	Katsuhiko Ogata, Modern Control Engineering Prentice-Hall of India, New Delhi.
2.	I.J. Nagarath and M. Gopal, Control system Engineering, New Age International (P) Ltd.
3.	Katsuhiko Ogata, State Space Analysis of Control Systems, Prentice Hall Inc, New Jersey.
4.	Benjamin C. Kuo and FaridGolnaraghi, Automatic Control Systems, 8th Edition, John Wiley & Sons.
Reference Books:	
1.	H. Khalil, Nonlinear Control systems, Prentice Hall Inc, New Jersey.
2.	Brogan W. L., Modern Control theory, Prentice Hall International, New Jersey.
3.	Jean-Jacques E, Slotine, Weiping Li, Applied Nonlinear Control, Prentice Hall Inc., New Jersey.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1209	PC	3	0	0	3	25	75	3 Hrs.
POWER SYSTEM SIMULATION LABORATORY-II								
Course Objectives:								
1.	To have knowledge on programming concept of neural network.							
2.	To understand the concept of genetic algorithm and its application in optimization.							
3.	To have the knowledge on fuzzy logic and design of fuzzy logic controllers.							
4.	To know the applications of AI Techniques in electrical engineering.							
5.	To use key computer simulation tools such as MATLAB for carrying out design experiments.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Acquire programming skills by utilizing neural networks.							K2
2.	Solve the different problems using training algorithms in neural networks.							K4
3.	Develop algorithms using genetic algorithm for optimization using MATLAB/SIMULINK software							K3
4.	Analyze and design fuzzy logic systems.							K4
EXPERIMENTS								
1.	ANN based Load Flow Analysis using Back Propagation Algorithm.							K3
2.	Economic load dispatch using Neural network.							K3
3.	State estimation of power systems using Neural network.							K3
4.	Contingency Analysis using Neural Network							K3
5.	Genetic Algorithm Program for maximization of sin(x) function using all basic operators.							K3
6.	Genetic Algorithm for Economic Load Dispatch of Thermal Units including losses (Use B-coefficients for computation of losses).							K3
7.	Simulation and verification of fuzzy Logic experiments using fuzzy logic trainer.							K3
8.	Single area and Two area Load Frequency Control using Fuzzy Logic Toolbox.							K3
9.	Fuzzy Logic based small signal stability analysis							K3
10.	Simulation of HVDC systems							K3
Reference Books:								
1.	Power Generation, Operation and Control - by A.J.Wood and B.F.Wollenberg, 2 nd edition, John wiley& sons Inc. 1984.							
2.	Power system operation and control - PSR Murthy, 1 st edition, B.S publication. 2008.							
3.	K. R. Padiyar, ‘High Voltage Direct Current Transmission’, Third Edition, New Age Publishers, New Delhi							
4.	Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications by RajasekharanandPai – PHI Publication							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1210	PC	3	0	0	3	25	75	3 Hrs.
POWER ELECTRONICS FOR RENEWABLE INTEGRATION LABORATORY								
Course Objectives:								
1.	To study and understand the different converters and inverters for single and three phase loads.							
2.	To study and understand the different renewable energy systems.							
3.	To study the characteristics of IGBT, MOSFET & GTO's. and their gate drive circuits							
4.	To analyse the Performance of Grid connected and Standalone Solar and wind Power System.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Evaluate the theoretical concepts of renewable energy systems							K3
2.	Demonstrate the modeling of PWM controled converters							K2
3.	Understand the working of IGBT, MOSFET & GTO's. and their gate drive circuits.							K3
4.	Compare the Performance of Grid connected and Standalone Solar and wind Power System..							K2
EXPERIMENTS								
1.	Study of 1- ϕ dual converter.							K3
2.	Study of 3- ϕ inverter with 120o and 180o mode of operation.							K3
3.	Study of 5-level cascaded H-bridge inverter.							K3
4.	Determination of input p.f. and harmonic factor for 3- ϕ full converter (Inductive load).							K3
5.	Study of 3-level NPC inverter.							K3
6.	Experiment on “VI-Characteristics and Efficiency of 1kWp Solar PV System”.							K3
7.	Experiment on “Shadowing effect & diode based solution in 1kWp Solar PV System”.							K3
8.	Experiment on Performance assessment of Grid connected and Standalone 1kWp Solar Power System.							K3
9.	Experiment on Performance assessment of micro Wind Energy Generator.							K3
10.	Experiment on Performance Assessment of Hybrid (Solar-Wind) Power System.							K3
11.	Study the characteristics of IGBT, MOSFET & GTO's.							K3
12.	Design of gate drive circuits for IGBT & MOSFET's.							K3
Reference Books:								
1.	Power Converter Circuits – William Shepherd & Li Zhang-Yes Dee Publishing Pvt Ltd.							
2.	Pulse Width Modulation: Analysis and Performance in Multilevel Inverters, Satish Kumar Peddapelli.First Edition,2017 Walter de Gruyter GmbH, Berlin/Boston.							
3.	Solar Energy-Fundamental Design, modelling and application by G N Tiwary: Narosa Publishing house, New Delhi							
4.	Renewable energy, Power for a sustainable future. By Godfrey Boyle: Oxford University press U.K.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS1211	MP	0	0	4	2	100	--	3 Hrs.
MINI PROJECT WITH SEMINAR								
For Mini Project with Seminar, a student under the supervision of a faculty member, shall collect the literature on a topic and critically review the literature and submit it to the department in a report form and shall make an oral presentation before the Project Review Committee consisting of Head of the Department, supervisor/mentor and two other senior faculty members of the department. For Mini Project with Seminar, there will be only internal evaluation of 100 marks. A candidate has to secure a minimum of 50% of marks to be declared successful.								

Code	Category	L	T	P	C	I.M	E.M	Exam
#AC-2	AC	2	0	0	0	0	0	--
AUDIT COURSE-2								
List of Audit Courses and their Syllabi are mentioned in the First Semester Syllabus. The students can opt any one course for AC 2 from the list mentioned in first semester by not opting the course which is already taken for AC 1								



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

M.TECH (POWER SYSTEM & AUTOMATION)

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

(With effect from **2019-2020** Admitted Batch onwards)

III-SEMESTER

Code No.	Name of the Subject	Category	C	L	T	P	Internal Marks	External Marks	Total Marks
#PE-V	Program Elective – V	PE	3	3	0	0	25	75	100
#Open Elective	Open Elective	OE	3	3	0	0	25	75	100
M19PS2106	Dissertation-I / Industrial Project	PR	10	0	0	20	50	50	100
Total			16	6	0	20	100	200	300

	Course Code	Course
#PE-V/ MOOCS	M19PS2101	Power System Automation.
	M19PS2102	Restructured Power System
	M19PS2103	Flexible AC Transmission systems
	M19 PS 2104 (MOOCS-I)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course
#OE-I/ MOOCS	#OE-I	Students have to choose one open elective course offered by departments other than the parent department. List of open Electives offered by other departments are enclosed.
	M19 PS 2105 (MOOCS-II)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course

OPEN ELECTIVES OFFERED TO OTHER DEPARTMENTS	
M19PS2107	Electric And Hybrid Vehicles
M19PS2108	Energy From Waste
M19PS2109	Energy Management and Auditing

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS2101	PE	3	0	0	3	25	75	3 Hrs
POWER SYSTEM AUTOMATION								
(Programme Elective-V)								
Course Objectives:								
S.No	Objective							
1	To familiarize the students with the basics of Power System Automation,Building blocks.							
2	To learn about Supervisory Control And Data Acquisition(SCADA) System,Remote Terminal Units(RTU), Master Stations etc							
Course Outcomes:								
S.No	Students are able to							Knowledge level
1	Understand the concepts of power system automation.							K2
2	Understand the components of SCADA systems.							K2
3	Comprehend the RTU, IED and other components of automation systems							K3
4	Understand the transfer of signals from the field to an operator control terminal.							K2
5	Design an interoperable powers automation system.							K5
SYLLABUS								
UNIT-I: (10 HOURS)	Evolution of Automation systems, History of Power system automation, Supervisory Control And Data Acquisition(SCADA) Systems, Components of SCADA systems, SCADA Applications, SCADA in power systems, SCADA basic functions, SCADA application functions in Generation, Transmission and Distribution.							
UNIT-II: (10 HOURS)	Advantages of SCADA in Power Systems, The Power system ‘Field’, Types of data &signals in the Power system, Flow of Data from the field to the SCADA Control center. Building blocks of SCADA systems, Classification of SCADA systems.							
UNIT-III: (10 HOURS)	Remote Terminal Unit (RTU), Evolution of RTUs, Components of RTU, Communication, Logic, Termination and Test/HMI Subsystems, Power supplies, Advanced RTU Functionalities.							
UNIT-IV: (10 HOURS)	Intelligent Electronic Devices (IEDs), Evolution of IEDs , IED functional block diagram, The hardware and software architecture of IED, IED Communication subsystem, IED advanced functionalities, Typical IEDs, Data Concentrators and Merging Units, SCADA Communication Systems.							
UNIT-V: (10 HOURS)	Master Station, Master station software and hardware configurations, Server systems inthe master station, Small, medium and large master station configurations, GlobalPositioning Systems, Master station performance, Human Machine Interface (HMI), HMIcomponents, Software functionalities, Situational awareness, Case studies in SCADA.							
Text Books:								
1	Mini S. Thomas, John D McDonald, Power Systems SCADA and Smart Grid, CRC Press, Taylorand Francis.							
2	Electric Power Substation Engineering John D. Mc Donald CRC Press, Taylor and Francis							
3	Control and Automation of Electrical Power Distribution systems, James Northcote- Green, RWilson, CRC Press, Taylor and Francis.							
4	Electric Power Distribution, Automation, Protection and Control, James Momoh, CRC press,Taylor and Francis.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS2102	PE	3	0	0	3	25	75	3 Hrs
RESTRUCTURED POWER SYSTEMS								
(Programme Elective-V)								
Course Objectives:								
1	To provide in-depth understanding of operation of deregulated electricity market systems.							
2	To examine typical issues in electricity markets and how these are handled world –wide in various markets.							
3	To enable students to understand the importance of ISO and effect of ancillary service in deregulation.							
4	To enable students to know the importance of unit commitment and Ancillary services.							
Course Outcomes:								
S.No	Students are able to							Knowledge level
1	Understand of operation of deregulated electricity market systems.							K2
2	Identify the Typical issues in electricity markets and importance of the Independent System Operator in Deregulation.							K3
3	Analyze various types of electricity market operational and control issues using new mathematical models.							K4
4	Understand the importance of Unit commitment and Ancillary services in Deregulation.							K2
SYLLABUS								
UNIT-I: (10 HOURS)	Deregulation, Need and conditions for deregulation, Different Entities in Deregulated Electricity Markets, Problems with Deregulating Electricity, Background to Deregulation and the Current Situation around the World, Benefits from a Competitive Electricity Market, After-Effects of Deregulation.							
UNIT-II: (10 HOURS)	Market Structure and Operation, Spot market, forward markets and settlements. Review of Concepts marginal cost of generation. Different structure model like Monopoly model, Purchasing agency model, wholesale competition model, Retail competition model.							
UNIT-III: (10 HOURS)	Role of the Independent System Operator (ISO), Operational Planning Activities of ISO, Operational Planning Activities of a Genco, Genco in Pool Markets, Genco in Bilateral Markets, Market Participation Issues, Unit Commitment in Deregulated Environment, Competitive Bidding.							
UNIT-IV: (10 HOURS)	Power exchange (PX), Market clearing price (MCP,) Market operations, Market power, Transmission pricing, contract path method, MW-mile method, LMP, Congestion management methods- market splitting, counter-trading, effect of LMP on Congestion.							
UNIT-V: (10 HOURS)	Ancillary Services and System Security in Deregulation, Classifications and definitions, Ancillary Services Management in Various Countries, Reactive Power as an Ancillary Service.							
Text Books:								
1	Power System Economics: Designing markets for electricity – IEEE Press &wiley inter science, Steven Stoft,2002							
2	Market operations in electric power systems - M. Shahidehpour, H. Yamin and Z. Li,Wiley,2002							
3	Operation of restructured power systems - K. Bhattacharya, M.H.J. Bollen and J.E. Daalder,							

	Springer,Kluwer Academic Publishers 2001
Reference Books:	
1	Fundamentals of power system economics - S. Kirschen and G. Strbac, Wiley.
2	Optimization principles: Practical Applications to the Operation and Markets of the Electric Power Industry - N. S. Rau, IEEE Press series on Power Engineeirng. Competition and Choice in Electricity - Sally Hunt and Graham Shuttleworth, Wiley
Web links	
1	https://nptel.ac.in/courses/108101005/3

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS2103	PE	3	0	0	3	25	75	3 Hrs
FLEXIBLE AC TRANSMISSION SYSTEMS								
(Programme Elective-V)								
Course Objectives:								
1	To learn the active and reactive power flow control in powersystem							
2	To understand the need for staticcompensators							
3	To develop the different control strategies used forcompensation							
Course Outcomes:								
S.No	Students are able to							Knowledge level
1	Acquire knowledge about the fundamental principles of Passive and Active Reactive Power Compensation Schemes at Transmission level in PowerSystems.							K3
2	Learn various Static VAR Compensation Schemes like Thyristor/GTOControlled.							K3
3	Explainthe need of Reactive Power Systems, PWM Inverter based Reactive Power Systems and theircontrols.							K3
4	Analyze the effect of Voltage and Phase Angle Regulators on power system stability.							K3
SYLLABUS								
UNIT-I: (10 HOURS)	Introduction to FACTS - Concept, power flow and stability, basic theory of line compensation; Thyristor controlled and converter-based FACTS controller - Power Electronic Controllers: Basic types of FACTS controllers, benefits from FACTS Controllers.							
UNIT-II: (10 HOURS)	Static Shunt Compensators: Midpoint voltage regulation; Variable impedance type and switching converter type static Var generators, SVC and STATCOM - TCR, TSC, V-I and V-Q characteristics, system stability, Static versus passive VAR compensator, Comparison between SVC and STATCOM.							
UNIT-III: (10 HOURS)	Static Series Compensators: Concept of series compensation, voltage stability, variable impedance type series compensators, GCSC, TSSC, TCSCand SSSC, control techniques, control range and VA rating.							
UNIT-IV: (10 HOURS)	SSR and its damping, Unified PowerFlowController, CircuitArrangement, Operation and control ofUPFC.Basic Principle of P and Q control- Independent real and reactive power flow control-Applications.–Interline Power Flow Controller (IPFC) – Operation and control							
UNIT-V: (10 HOURS)	Static Voltage and Phase Angle Regulators: Power flow control, TCVR and TCPAR, improvement of transient stability-Modelling of FACTS devices, optimization of FACTS,transient and dynamic stability enhancement							
Text Books:								
1	K R Padiyar, “FACTS Controllers in Power Transmission and Distribution”, New Age International Publishers,2007. Second edition							
2	N.G. Hingorani, L. Gyugyi, “Understanding FACTS: Concepts and Technology of Flexible AC Transmission Systems”, IEEE Press Book, Standard Publishers and Distributors, Delhi,1st ed,2001.							
3	X P Zhang, C Rehtanz, B Pal, “Flexible AC Transmission Systems- Modelling and Control”.SpringerVerlag. Berlin.1st ed. 2006 edition							

Reference Books:	
1	Power Quality Enhancement Using Custom Power Devices – Power Electronics and Power Systems, Gerard Ledwich, Arindam Ghosh, Kluwer Academic Publishers, 1st ed, 2002.
2	FACTS Controllers: applications of static converters in power systems- Vijay K.Sood- Springer publishers
Web links	
1	https://nptel.ac.in/courses/108107114/

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 PS 2104	PE	3	0	--	3	25	75	3 Hrs.
MOOCS-I								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course								

Code	Category	L	T	P	C	I.M	E.M	Exam
	OE	--	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE								
Students have to choose one open elective course offered by departments other than the parent departments. List of open Electives offered by other departments are given below.								

Offered from	Course Code	Course Name	Offered to
CIVIL ENGINEERING	M19 ST 2107	Construction Management	CST, CS, PSA,IT & CAD/CAM
	M19 ST 2108	Green Technology	
	M19 ST 2109	Analysis of Offshore Structures	
COMPUTER SCIENCE & ENGINEERING	M19 CST 2106	Python Programming	ST, CS, PSA & CAD/CAM
	M19 CST 2107	Artificial Intelligence	
	M19 CST 2108	Advanced Data structures	
ELECTRONICS & COMMUNICATION ENGINEERING	M19 CS 2107	Signals and systems	ST, CST, PSA, IT & CAD/CAM
	M19 CS 2108	Principles of Communication	
	M19 CS 2109	Image and video Processing	
INFORMATION TECHNOLOGY	M19IT2108	Web Technologies	ST, CS, PSA & CAD/CAM
	M19IT2109	Internet of Things	
	M19IT2110	Machine Learning	
MECHNAICAL ENGINEERING	M19CAD 2107	Operations Research	ST, CST, CS, PSA & IT
	M19CAD 2108	Nano Technology	
	M19CAD 2109	Product Design & Manufacturing	
SCIENCE & HUMANITIES	M19BS2101	Management and Organisational Behaviour	ST, CST, CS, PSA, IT & CAD/CAM

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 PS 2105	OE	3	0	--	3	25	75	3 Hrs.
MOOCS-II								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course								

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS2106	PR	0	0	20	10	50	50	3 Hrs.
DISSERTATION-I/INDUSTRIAL PROJECT								
The Student has to register for Dissertation-I / Industrial project in III semester. Student has to submit, in consultation with his project supervisor, the title, objective and plan of action of his project work for approval. The student can initiate the Project work, only after obtaining the approval from the Project Review Committee (PRC). Continuous assessment of Dissertation-I during the III-Semester will be monitored by the PRC. Dissertation-I/ Industrial Project is evaluated for 50 internal marks and 50 external marks. Internal marks 50 awarded by Project Guide and PRC jointly based on continuous assessment consisting of two seminars based on Dissertation work-I. External marks 50 awarded by External Examiner, Supervisor and Head of the Department jointly based on a review and Viva voce on Dissertation work-I.								

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R19)
M.TECH (POWER SYSTEM & AUTOMATION)
DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

(With effect from **2019-2020** Admitted Batch onwards)

IV-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Internal Marks	External Marks
M19PS2201	Dissertation-II / Industrial Project	PR	16	0	0	32	--	100

Code	Category	L	T	P	C	I.M	E.M	Exam
M19PS2201	PR	0	0	32	16	--	100	3 Hrs.
DISSERTATION-II/INDUSTRIAL PROJECT								
The student has to continue his/her work from Dissertation-I / Industrial project to complete Dissertation-II in IV semester. Continuous assessment of Dissertation-II during IV-Semester will be monitored by the PRC. Dissertation-II is evaluated for 100 external marks based on Review and Viva Voce. Review and Viva-Voce examination shall be conducted by a board consisting of the Supervisor, Head of the Department and the examiner who adjudicated the Thesis. The Board shall jointly report the candidate's work for 100 marks. If the report of the Viva-Voce is unsatisfactory (ie, < 50 marks), the candidate shall retake the Viva-Voce examination only after three months. If he fails to get a satisfactory report at the second Viva-Voce examination, the candidate has to reregister for the project and complete the project within the stipulated time after taking the approval from the College.								